

SYE6047: QUALITY IMPROVEMENT: SYSTEMS AND METHODOLOGIES

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Quality Improvement: Systems and Methodologies

Subject Code

SYE - Systems Engineering

Course Number

6047

Academic Unit

Systems Engineering (SYE)

College/School

College of Engineering (EG)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

SEEM3062 - Quality Engineering I (offered until 2016/17) or SEEM3102 - Quality Engineering (offered until 2020/21) or ADSE3102 Quality Engineering (offered until 2023/24) or SYE3102 Quality Engineering

Equivalent Courses

SEEM6047 Quality Improvement: Systems and Methodologies (offered until 2021/22) / ADSE6047 Quality Improvement: Systems and Methodologies (offered until 2023/24)

Exclusive Courses

Nil

Part II Course Details

Abstract

To provide students with a basic understanding of the approaches, systems and statistical techniques to assess and improve product/service quality in a manufacturing/ service organization, and to equip students with modern quality improvement tools and appreciation of their implementation issues in product/ service.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Define the various dimensions of quality in product and service realization.	10	x	x	
2	Apply the concepts and principles of total quality management and six sigma quality philosophy in developing company wide quality systems.	20			
3	Apply structured and data-driven approach to identify quality problem, define and measure key process steps and inputs, and identify potential root causes of the problem.	30	x	x	
4	Apply statistical tools to validate cause and effect relationship between process inputs and outputs, derive and validate improvement solutions to address the root causes of the problem.	30	x	x	
5	Discuss the framework and associated issues of implementing and monitoring of quality improvement projects.	10	x		

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Large Class Activities	Learning through teaching is primarily based on lectures. Mini-lectures and small-group exercises will be used to facilitate conceptual understanding and industrial applications of various statistical tools and techniques.	1, 2, 3, 4, 5	27 hours/sem
2	Group Term Paper	The team-based term paper provides students with the opportunities to conduct literature-based study of published or real life applications of Six Sigma quality philosophy in various manufacturing and non-manufacturing industries.	3, 4	6 hours/sem
3	Exercises	The team-based exercises enable students to design, conduct and analyze factorial experiments and gage R&R study in solving practical problems.	4	6 hours/sem

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("-" for nil entry)	Allow Use of GenAI?
1	Large Class Activities and Exercises	1, 2, 3, 4	30	nil	No
2	Group Term Paper and Presentation	3, 4	20	nil	No

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

30

Minimum Examination Passing Requirement (%)

30

Assessment Rubrics (AR)

Assessment Task

Exercises (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The exercises assess students' ability to design, conduct and analyze factorial experiments and gage R&R study with statistical package. Interpretations of the numerical results and their practical implications are particularly sought for.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Group Term Paper and Presentation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Students' ability to solicit, digest, and organize materials of real life Six Sigma applications are assessed through written report and oral presentation. Key activities results and possible financial impact of the quality problem solving cycle are required.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Examination questions are designed to assess student's level of achievement of the intended learning outcomes, with balanced emphasis placed on both conceptual understanding of the statistical tools introduced and practical applications of the various quality improvement systems and methodologies.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Exercises (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The exercises assess students' ability to design, conduct and analyze factorial experiments and gage R&R study with statistical package. Interpretations of the numerical results and their practical implications are particularly sought for.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate/Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Group Term Paper and Presentation (for students admitted from Semester A 2022/23 to Summer Term 2024)

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Additional Information for AR

The quiz(s), laboratory report and the case study will be numerically-marked, while examination will be numerically-marked and grades-awarded accordingly. The areas of achievement to be assessed for each activity are summarized as below.

Part III Other Information**Keyword Syllabus**

- Product/service quality definition and dimensions
- Six Sigma and DMAIC methodology
- Measure and analysis: measurement system analysis, rolled throughout yield, SIPOC analysis, Gage R&R, confidence interval, hypothesis testing;
- Improve and control: correlation and regression, analysis of variance, DOE (design of experiments), randomized blocks, BIBD, full factorial experiments, designed matrix experiments, control planning and application.
- Quality improvement implementation framework, team dynamics, improvement project planning

Reading List**Compulsory Readings**

Title	
1	Nil

Additional Readings

Title	
1	H S Gitlow and D M Levine, Six sigma for green belts and champions: foundation, DMAIC, tools, cases and certification, Pearson/PrenticeHall, 2005
2	Summer, Donna C S, Six Sigma: Basic Tools and Techniques, Pearson/PrenticeHall, 2007
3	Greg Brue, Six Sigma for Managers, McGraw-Hill, 2002
4	Gryna, Quality Planning and Analysis, 4th ed., Mc-Graw Hill, 2001
5	D.C. Montgomery, Design and Analysis of Experiments, 7th ed., Wiley, 2008
6	D.C. Montgomery, Introduction to Statistical Quality Control, 5th ed., Wiley, 2005
7	Hines & Montgomery, Probability and Statistics in Engineering and Management Science, 3rd ed., Wiley, 1990
8	G. S. Peace, Taguchi Methods : A Hands-On Approach, Addison Wesley, 1993